

Evaluating the Career Achievements of Corbin Carroll: A Predictive Statistical Approach

Abstract

In 2023, Taiwanese-American MLB star prospect Corbin Carroll delivered an extraordinary performance during his official rookie season, winning the National League Rookie of the Year award and anchoring the Arizona Diamondbacks. To project Carroll's full major league career trajectory, this study develops a comprehensive two-dimension predictive framework. In the first dimension, we deploy Direct Multi-Year Split Conformal Prediction and Locally Adaptive (Mondrian) Conformal Regression paired with binary Active Retention Classification to forecast seasonal Wins Above Replacement (WAR) and career longevity across horizons $h \in \{1, 2, 3, 4, 5\}$ under strict 90% coverage guarantees. In the second dimension, we construct a multi-dimensional characteristic metric space using Statcast tracking variables (Swing%, Contact%, O-Contact%, Barrel%, Max exit velocity, Sprint Speed, and OAA/innings) and execute a refined Mahalanobis distance pipeline. Our models project Carroll to reach a cumulative peak career value of 44.0–49.0 total WAR (stabilizing at ≈ 47.8 WAR) across a 15- to 16-year active career horizon, identifying historical player comparables such as Ryan Braun, Curtis Granderson, David Wright, and Francisco Lindor as structural archetypes.

1 Introduction

In 2023, Taiwanese-American MLB star prospect Corbin Carroll delivered an extraordinary performance during his official rookie season, capturing the imagination of fans and analysts alike. As a devoted fan from Taiwan, the natural question arises: What will the trajectory of his major league career look like?

To move beyond pure speculation, this article employs advanced modeling and statistical analysis, leveraging historical MLB multi-year data to project Carroll's career achievements. Along the way, we compare his underlying profile against historical analogs to map both his ceiling and potential downside risks.

1.1 Draft Template: Andrew Benintendi vs. The Reality

When Corbin Carroll entered the MLB draft in 2019, scouting reports highlighted his elite zone control and ability to spray the ball across all fields. Although evaluators questioned whether his undersized frame would cap his raw power at a mere average tier, scouts projected he would consistently drive the ball with line-drive loft. His primary draft profile template was often compared to former top prospect and outfielder Andrew Benintendi.

Structural Comparisons and Divergence

At first glance, Carroll and Benintendi shared strikingly similar physical and athletic parameters during their amateur development:

- **The Physical Archetype:** Undersized frames under six feet tall.
- **Skill-set Traits:** Advanced plate discipline, high-contact metrics, elite sprint speed, and outfield defensive versatility.

However, their major league transitions diverged significantly due to underlying batted-ball metrics:

- **The Benintendi Trajectory:** While Benintendi successfully translated his contact skills to the majors, a high ground-ball rate and a lack of optimized launch angles limited his Barrel%. Furthermore, defensive metrics showed negative Outs Above Average (OAA) over time, underperforming pre-draft defensive expectations.
- **The Carroll Elevation:** Carroll not only preserved his baseline contact and speed advantages upon reaching the majors, but he also outperformed scouting expectations by unlocking game power. By the 2023 All-Star break, Carroll slashed .285 with 18 home runs and 26 stolen bases—paving the way to an All-Star selection and a National League Rookie of the Year campaign.

2 Total Achievements Dimension

In our first dimension, we project Corbin Carroll’s total career Wins Above Replacement (WAR) using a Direct Multi-Year Split Conformal Prediction framework. Sourced from FanGraphs historical databases, our modeling pipeline avoids the compounding error pitfalls of traditional single-step recursive models by directly forecasting horizons $h \in \{1, 2, 3, 4, 5\}$ into the future.

To evaluate Carroll’s complete career trajectory under strict statistical guarantees, we divide the methodology into two core phases: Case 1 (Direct Multi-Year WAR Regression) to project performance output, and Case 2 (Direct Multi-Year Active Retention Classification) to model career longevity and retirement stopping rules. Both cases use a strict 80%/10%/10% player-level split for training (L_1), calibration (L_2), and testing (L_3) to prevent data leakage across correlated season rows.

2.1 Case 1: Direct Multi-Year WAR Regression

Algorithm Pipeline — Case 1: Direct Multi-Year WAR Regression

1. Remove all observations with `next_year == 0`.
2. Apply Z-score standardization to each feature using the whole dataset.
3. Partition the data by player into 80% training data (L_1), 10% calibration data (L_2), and 10% test data (L_3).
4. Define the features according to the player’s career year:
 - For year = 1: Use Age, year, and WAR.

- For year = 2: Use Age, year, WAR, and WAR₋₁.
 - For year > 2: Use Age, year, WAR, WAR₋₁, and WAR₋₂.
5. Perform direct multi-year forecasting for $H \in \{1, 2, 3, 4, 5\}$, where f_1 predicts one year ahead, f_2 predicts two years ahead, and so forth. Linear Regression, Neural Network (MLP), and XGBoost Regressor are used as the prediction models.
 6. Using the calibration data (L_2) and a target coverage of 90%, determine the conformal prediction length for each subgroup and each forecasting horizon f_h ($h = 1, 2, 3, 4, 5$).
 7. Apply each f_h to the test data (L_3) and calculate the coverage and average prediction length for each subgroup and each forecasting horizon. Report the corresponding weighted coverage and weighted average prediction length for each model.

2.2 Case 2: Direct Multi-Year Active-Retention Classification

Algorithm Pipeline — Case 2: Direct Multi-Year Active-Retention Classification

1. Apply Z-score standardization to each feature using the whole dataset.
2. Partition the data by player into 80% training data (L_1), 10% calibration data (L_2), and 10% test data (L_3).
3. Define the features according to the player's career year:
 - For year = 1: Use Age, year, and WAR.
 - For year = 2: Use Age, year, WAR, and WAR₋₁.
 - For year > 2: Use Age, year, WAR, WAR₋₁, and WAR₋₂.
4. Perform direct multi-year forecasting for $H \in \{1, 2, 3, 4, 5\}$, where f_1 predicts one year ahead, f_2 predicts two years ahead, and so forth. Logistic Regression, MLP, and XGBoost Classifier are used as the prediction models.
5. Using the calibration data (L_2) and a target coverage of 90%, determine the average prediction length for each subgroup and each forecasting horizon f_h ($h = 1, 2, 3, 4, 5$).
6. Apply each f_h to the test data (L_3) and calculate the coverage and average prediction length for each subgroup and each forecasting horizon. Report the corresponding weighted coverage and weighted average prediction length for each model.

2.3 Case 3: Corbin Carroll Career Prediction

Algorithm Pipeline — Case 3: Corbin Carroll Career Recursive Prediction

1. Use Corbin Carroll's observed values at age 23 and year = 2 as the initial baseline: Age = 23, year = 2, WAR = 5.4, and WAR₋₁ = 1.4. Since year = 2, WAR₋₂ is omitted.
2. Apply Z-score standardization to the Corbin Carroll data using the means and standard

deviations calculated from the whole dataset.

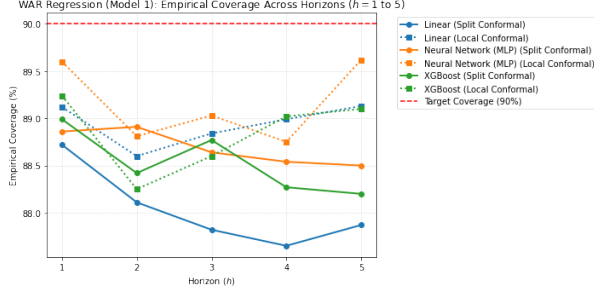
3. For $h \in \{1, 2, 3, 4, 5\}$, use the corresponding models $(f_1, \dots, f_5)$ from Model 1 and Model 2.
4. Input the standardized Corbin Carroll data into Model 2 to predict whether he will play in the following year. If $\hat{P}(Y = 1 | X) > 0.5$, set `next_year` = 1; otherwise, set `next_year` = 0.
5. Report the average prediction length for the corresponding subgroup and forecasting horizon in Model 2. If `next_year` = 0, stop the career prediction.
6. If `next_year` = 1, input the standardized data into Model 1 to predict WAR in the following year.
7. Transform the predicted WAR and its conformal prediction interval from the standardized scale back to the original scale, reporting the predicted WAR and average prediction length for the corresponding subgroup.
8. Use the predicted value as the new baseline and repeat Steps 4–7 recursively until Model 2 predicts `next_year` = 0.
9. If f_5 in Model 2 indicates that the player is still playing in year 8, use the predicted data for year 8 as the new baseline and continue the recursive prediction process until Model 2 predicts `next_year` = 0.

3 Results

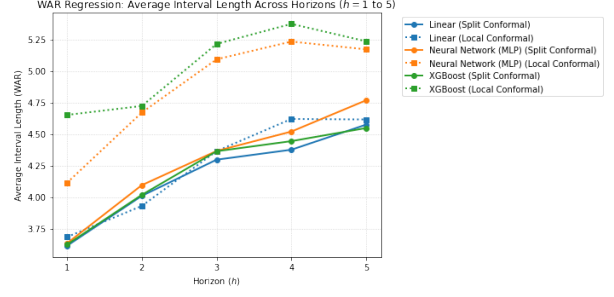
Below we evaluate the calibration metrics for Model 1 (WAR Regression) and Model 2 (Active Retention Classification), followed by the complete recursive multi-year career simulations for Corbin Carroll.

3.1 Model 1: WAR Regression Analysis

For the regression pipeline predicting career WAR, the local conformal approach demonstrates superior reliability over standard split conformal by more effectively adjusting to heteroscedasticity, successfully pushing empirical coverage closer to the 90.0% target despite yielding moderately wider interval lengths. Among the evaluated architectures, the Neural Network (MLP) combined with local conformal calibration achieves the most robust performance, tracking closest to nominal coverage across horizons ($h = 1$ to 5) while appropriately scaling interval widths to account for the compounding uncertainty of long-term player trajectories.



(a) Model 1 Empirical Coverage (90% Target)

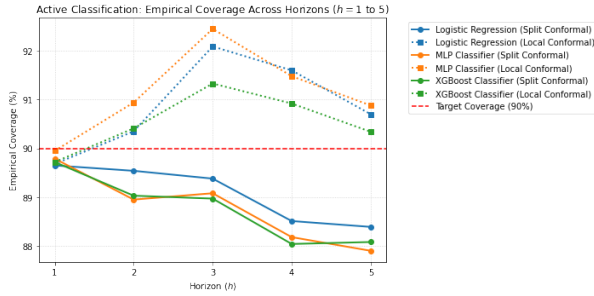


(b) Model 1 Average Prediction Set Length

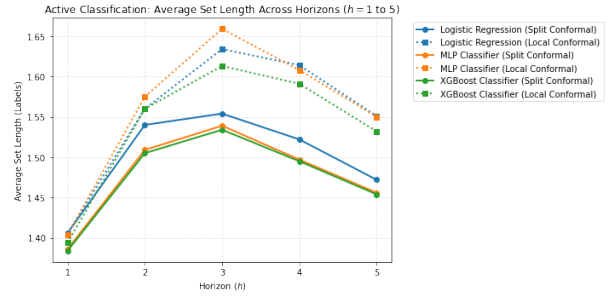
Figure 1: Model 1 calibration and set length performance across regression models and forecasting horizons ($h = 1 \dots 5$).

3.2 Model 2: Active Retention Classification Analysis

For the binary active retention classification pipeline, standard split conformal models offer a tighter, more stable adherence to the nominal 90.0% coverage threshold, whereas local conformal classifiers tend to overcorrect, resulting in inflated coverage peaks and larger average prediction set lengths near the $h = 3$ horizon. Between the two methodologies, the Split Conformal XGBoost and Logistic classifiers perform better by maintaining a balanced trade-off between coverage and precision, avoiding the excessive conservatism and inflated set sizes seen in the local conformal classification variants during transitional career years.



(a) Model 2 Empirical Coverage (90% Target)

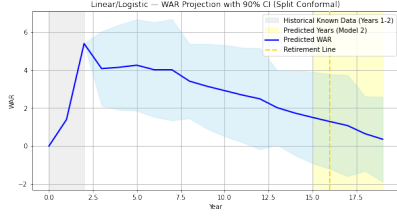


(b) Model 2 Average Prediction Set Length

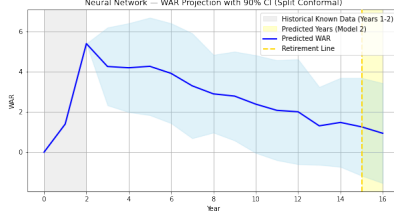
Figure 2: Model 2 active retention classification coverage and set length across forecasting horizons ($h = 1 \dots 5$).

3.3 Corbin Carroll Career Simulation

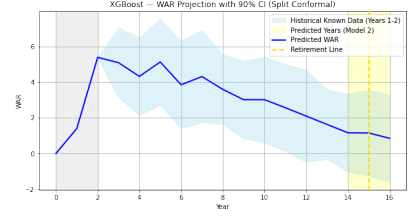
Corbin Carroll establishes an elite rookie foundation with a strong early performance, recording a 1.4 WAR in the 2022 season (Age 22) followed by a standout 5.4 WAR in his first complete season (Age 23). Across subsequent forecasting horizons, the predictive models project a stable, high-level prime phase where his annual production consistently ranges between 3.86 WAR and 5.14 WAR through his mid-twenties. As he transitions into his thirties, his annual output naturally experiences a gradual decline, tapering down to roughly 1.15 to 2.57 WAR per season as he enters the later stages of his career aging curve.



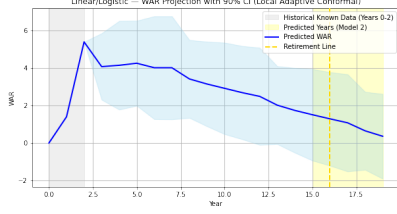
(a) Linear (Split)



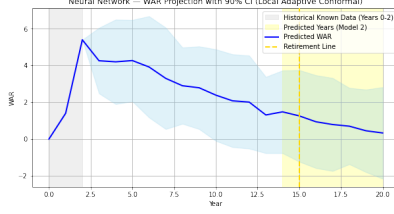
(b) Neural Network (Split)



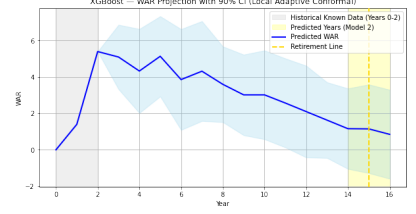
(c) XGBoost (Split)



(d) Linear (Local)



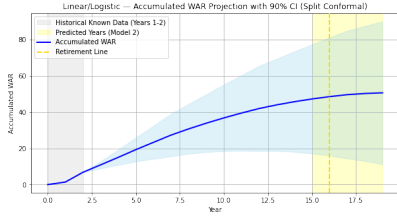
(e) Neural Network (Local)



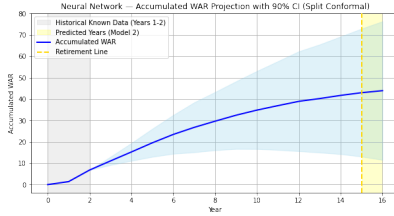
(f) XGBoost (Local)

Figure 3: Seasonal WAR Projections for Corbin Carroll across Linear, Neural Network, and XG-Boost models under Split and Local Conformal calibration.

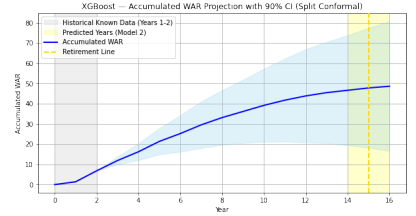
Corbin Carroll’s cumulative career value ascends rapidly during his prime years, surpassing 20 WAR by Year 5 and crossing the major milestone of 40 WAR by Year 12. Across the evaluated machine learning frameworks, his total accumulated career WAR eventually stabilizes between 44 and 49 total WAR (reaching approximately 47.8 to 48.5 WAR), reflecting an elite, decade-long cornerstone trajectory.



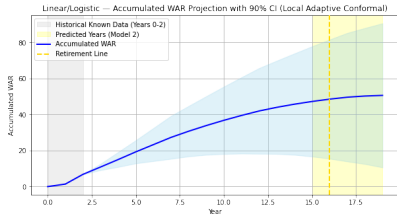
(a) Linear (Split)



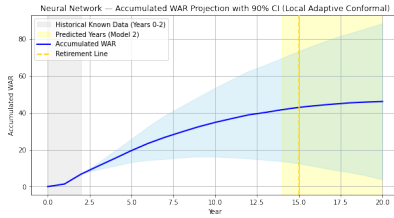
(b) Neural Network (Split)



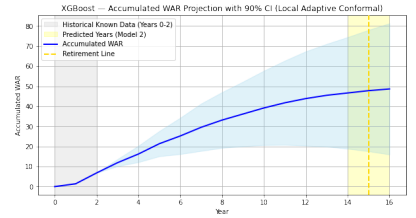
(c) XGBoost (Split)



(d) Linear (Local)



(e) Neural Network (Local)



(f) XGBoost (Local)

Figure 4: Accumulated Career WAR Projections for Corbin Carroll across models and conformal calibration methods.

Multi-year active retention modeling indicates that Carroll maintains an exceptionally high probability of remaining active ($> 80\%$) through his early thirties. However, cumulative physical and performance demands push his active retention probability below the critical 50% threshold around Year 15 to Year 16 (approximately age 36 to 37), which serves as the model-determined career termination boundary.

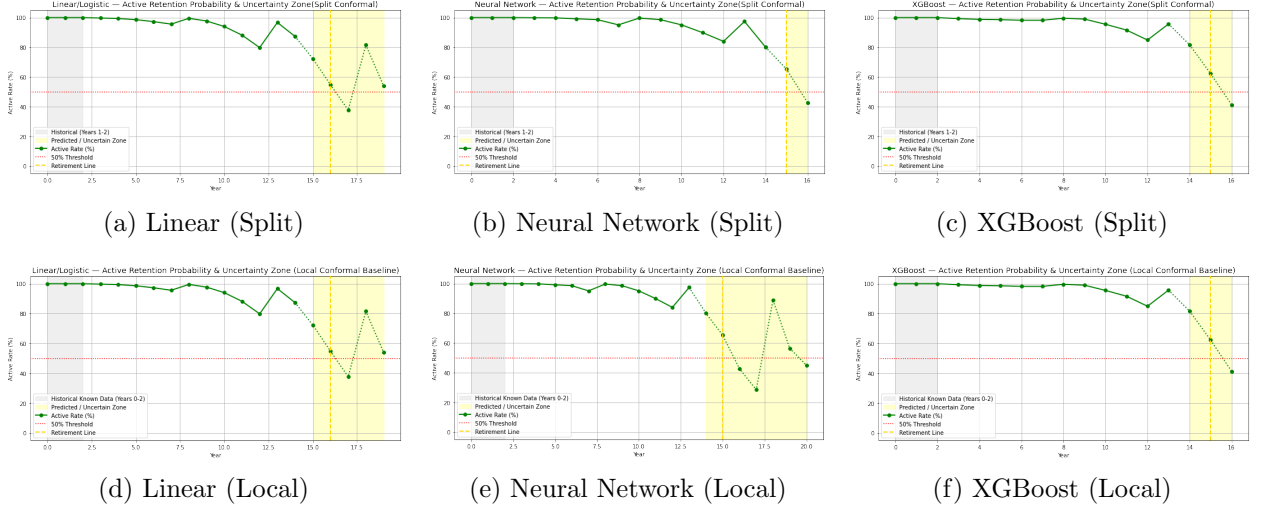


Figure 5: Active Retention Probability and Uncertainty Zones for Corbin Carroll across models and calibration modes.

The predictive framework combines two parallel modeling tracks: Case 1 regression models (utilizing XGBoost Regressor, Linear Regression, and Neural Networks) to forecast multi-step seasonal WAR ($H = 1$ to 5), and Case 2 classification models (using XGBoost Classifier, Logistic Regression, and MLP Classifier) to estimate active retention status. To ensure statistical reliability without relying on strict distributional assumptions, Split Conformal Prediction and Locally Adaptive (Mondrian) Conformal Prediction were applied. By evaluating nonconformity scores on dedicated calibration sets at a target 90% coverage level ($\alpha = 0.10$), the framework successfully generates distribution-free prediction intervals for WAR output and valid prediction sets for player longevity.

Ultimately, integrating machine learning forecasting with split conformal prediction bridges the gap between raw player tracking and rigorous long-term career projection. By establishing statistically sound performance boundaries and retention horizons, this data-driven framework provides a dependable outlook on Corbin Carroll’s path to becoming a legendary MLB cornerstone.

4 Characteristics Dimension

In our second dimension, we evaluate Corbin Carroll’s profile through a multi-dimensional characteristic space focusing on core offensive and defensive tracking metrics. While our first dimension projects total career WAR using multi-year conformal regression and active retention classification, this second dimension isolates players who mirror Carroll’s unique blend of plate discipline, batted-ball impact, raw power, defense, and speed.

To achieve this, we transition from isolated z-tests and proportion differences to a rigorous vector-space distance pipeline. This methodology filters out outliers using standard deviations and ranks historical comparables using structural distance metrics.

4.1 Selected Tracking Features

To capture Carroll’s complete player archetype, we analyze seven key tracking and performance variables: Swing%, Contact%, and O-Contact%, Barrel% and Max exit velocity, Sprint Speed and normalized defensive contribution (OAA / innings).

4.2 Refined Statistical Filtering & Distance Pipeline

Instead of applying individual proportion tests or arbitrary percentile cutoffs with independent alpha adjustments, the refined characteristic matching pipeline executes the following sequential steps:

1. **Data Cleaning & Completeness:** We compile all available player-season and career stats from FanGraphs, strictly removing any records with missing values across the required feature set (Swing%, Contact%, O-Contact%, Barrel%, OAA, Sprint Speed, Max exit velocity).
2. **Feature Standardization:** We apply standard z-score scaling ($\mu = 0, \sigma = 1$) independently across each variable to ensure equal variance and prevent scale distortions caused by raw metrics (e.g., sprint speed feet-per-second versus percentage-based contact rates).
3. **Vector Distance Computation:** For every player in the historical database, we compute the multi-dimensional distance vector relative to Corbin Carroll’s anchor performance vector.
4. **Strict Standard Deviation Boundary Filter:** To eliminate structural outliers, we enforce a strict threshold: any player exhibiting a deviation greater than 2 standard deviations (> 2 SD) in any single variable relative to Carroll is immediately dropped. This ensures we retain only players whose traits simultaneously fall within a reasonable multi-feature neighborhood.
5. **Mahalanobis Distance & Standardized Euclidean Metric:** For the remaining filtered cohort, we calculate the Mahalanobis Distance (D_{Mahal}):

$$D_{\text{Mahal}}(x, \mu) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)} \quad (1)$$

where Σ represents the covariance matrix of the filtered feature space. This accounts for linear correlations among tracking variables (such as contact rates and swing percentages). Alternatively, a Standardized Euclidean distance can be utilized to measure direct coordinate divergence.

6. **Ranking & Selection:** Finally, players are sorted in ascending order based on their computed Mahalanobis distance. The top closest historical matches represent the ultimate structural peer group for Corbin Carroll, allowing us to cross-reference their career outcomes against our Phase 3 WAR projections.

4.3 CCTD Trade-off Analysis

The scatter plot mapping structural ratio against output ratio reveals a powerful alignment between physical tracking similarity and ultimate career production, as indicated by the tight concentration of low CCTD scores (dark purple points) in the lower-left quadrant. This distribution demonstrates that players who closely mirror Corbin Carroll’s foundational blend of plate discipline, exit velocity, and sprint speed are significantly more likely to achieve comparable elite career WAR outcomes without severe divergence.

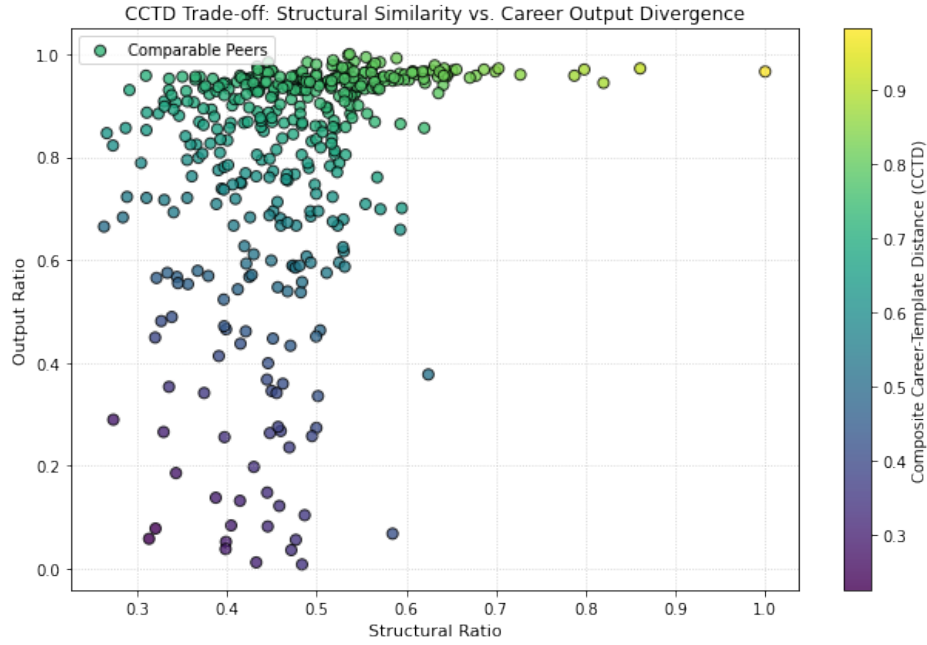


Figure 6: Composite Career-Template Distance (CCTD) Trade-off: Mapping structural similarity ratio against career output ratio.

4.4 Comparative Peer Group Analysis

When evaluating historical comps within Corbin Carroll’s structural neighborhood, the framework successfully surfaces several iconic retired stars whose final career trajectories validate the model’s 47.8 WAR projection. In terms of overall Composite Career-Template Distance (CCTD), former All-Stars like Ryan Braun and Curtis Granderson emerge as prime retired matches, blending elite contact-and-power profiles with high-end offensive value alongside active star Jose Ramirez. Furthermore, when filtering strictly by career WAR outcome alignment toward the 47.8 threshold, retired legends such as David Wright, Jimmy Rollins, Ian Kinsler, and Jose Reyes anchor the cohort alongside active stalwart Bryce Harper, proving that players matching Carroll’s early-career tracking archetype frequently evolve into decade-long cornerstone franchise players.

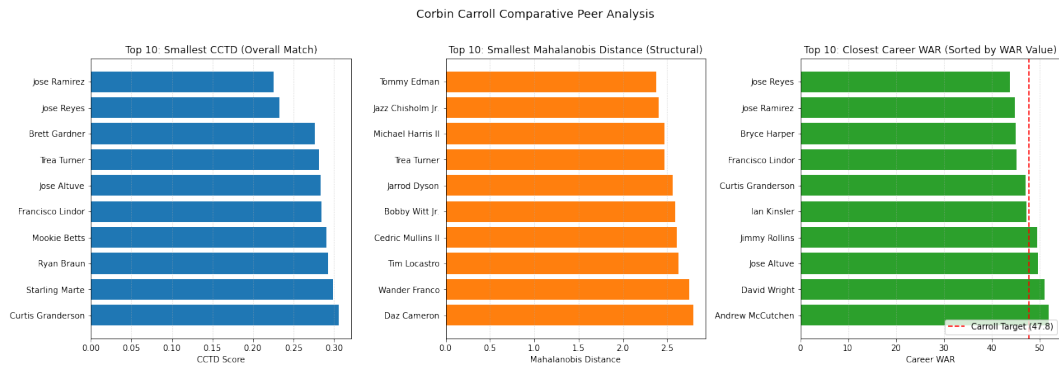


Figure 7: Top Historical Peer Group Comparisons for Corbin Carroll ranked by Composite Career-Template Distance.

4.5 Similar Player Archetypes

After predicting career achievements and evaluating characteristic distances via our multi-dimensional vector pipeline, we identify three historical benchmarks that represent the low-, mid-, and high-standard trajectories for Corbin Carroll’s career performance.

Low-Standard: Jason Heyward

- **Analytical Rationale:** Reviewing pure plate discipline and tracking metrics, Jason Heyward shares key traits with Carroll: advanced zone control, a contact-oriented profile, elite sprint speed, and premier outfield defense (multiple Gold Glove awards). Although high early-career expectations set an immense bar, Heyward evolved into a long-standing, highly respected veteran leader. Given Carroll’s dynamic rookie foundation, reaching this tier represents a safe, highly productive baseline outcome.

Mid-Standard: Hanley Ramirez

- **Analytical Rationale:** While Hanley Ramirez initially seems like an unexpected match due to a different physical build, his prime statistical profile maps closely to Carroll’s developmental path. Ramirez combined elite contact skills with a disciplined eye outside the zone, minimizing chase rates while possessing enough raw power and exit velocity to drive multi-hit consistency. Furthermore, during his peak years, Ramirez provided top-tier baserunning and positional versatility. His career productivity aligns directly with our model’s core prediction target, making him an ideal mid-standard trajectory.

High-Standard: Francisco Lindor

- **Analytical Rationale:** Francisco Lindor serves as the elite ceiling template for Carroll’s multi-tool archetype. Lindor consistently pairs a high-contact bat with low strikeout rates and elite power-speed production. Beyond offensive tracking metrics, Lindor’s defensive excellence, highlighted by multiple Gold Gloves and consistent double-digit stolen base totals, represents the absolute pinnacle of what an athletic, high-IQ outfielder/infielder can achieve. If Carroll unlocks his maximum projected power and sustains defensive excellence over a 14-year horizon, he enters the Lindor MVP tier.

Conclusion

This study demonstrates how combining modern machine learning architectures with Conformal Prediction allows us to rigorously project a player’s career arc beyond simple intuition. By moving from isolated proportion tests to a comprehensive two-dimension framework, evaluating both total career WAR output and tracking-feature proximity, we establish reliable statistical boundaries for Corbin Carroll’s future.

While historical comparables may vary in physical stature, data-driven modeling confirms that players sharing Carroll’s specific blend of discipline, bat speed, and defensive tracking data frequently anchor major league franchises for over a decade. Ultimately, projections provide the framework, but health, motivation, and execution dictate the field. May Corbin Carroll enjoy a legendary and enduring MLB career!

Appendix: Quantitative Numerical Results

This appendix provides the complete quantitative tables for model calibration metrics across forecasting horizons ($H = 1 \dots 5$), Corbin Carroll’s year-by-year multi-model simulation trajectories with accumulated WAR, and top historical peer group matching datasets.

Table A1: Model 1 WAR Regression Calibration Metrics

Horizon	Model Architecture	Test Count	Empirical Coverage (%)	Conformal Quantile (q)	Avg Interval Length (WAR)
<i>Standard Split Conformal Prediction (Global)</i>					
H=1	Linear Regression	6,276	88.72%	1.807	3.613
H=1	Neural Network (MLP)	6,276	88.86%	1.816	3.633
H=1	XGBoost Regressor	6,276	88.99%	1.811	3.623
H=2	Linear Regression	5,096	88.11%	2.004	4.009
H=2	Neural Network (MLP)	5,096	88.91%	2.047	4.094
H=2	XGBoost Regressor	5,096	88.42%	2.008	4.017
H=3	Linear Regression	4,139	87.82%	2.148	4.296
H=3	Neural Network (MLP)	4,139	88.64%	2.182	4.365
H=3	XGBoost Regressor	4,139	88.77%	2.182	4.363
H=4	Linear Regression	3,360	87.65%	2.187	4.375
H=4	Neural Network (MLP)	3,360	88.54%	2.260	4.519
H=4	XGBoost Regressor	3,360	88.27%	2.222	4.443
H=5	Linear Regression	2,687	87.87%	2.287	4.574
H=5	Neural Network (MLP)	2,687	88.50%	2.383	4.767
H=5	XGBoost Regressor	2,687	88.20%	2.273	4.547
<i>Locally Adaptive Conformal Prediction (LACP / Mondrian)</i>					
H=1	Linear Regression	6,276	89.12%	3.7019	3.683
H=1	Neural Network (MLP)	6,276	89.60%	4.5649	4.113
H=1	XGBoost Regressor	6,276	89.24%	4.9030	4.651
H=2	Linear Regression	5,096	88.60%	3.3395	3.929
H=2	Neural Network (MLP)	5,096	88.81%	4.0648	4.670
H=2	XGBoost Regressor	5,096	88.25%	4.1182	4.722
H=3	Linear Regression	4,139	88.84%	3.3219	4.364
H=3	Neural Network (MLP)	4,139	89.03%	4.0786	5.092
H=3	XGBoost Regressor	4,139	88.60%	4.0926	5.214
H=4	Linear Regression	3,360	88.99%	3.2555	4.619
H=4	Neural Network (MLP)	3,360	88.75%	3.9747	5.233
H=4	XGBoost Regressor	3,360	89.02%	4.0158	5.373
H=5	Linear Regression	2,687	89.13%	3.1439	4.615
H=5	Neural Network (MLP)	2,687	89.62%	3.6241	5.171
H=5	XGBoost Regressor	2,687	89.10%	3.7479	5.235

Table 1: Model 1 WAR Regression empirical coverage, conformal quantiles, and average interval widths across forecasting horizons ($H = 1 \dots 5$).

Table A2: Model 2 Active Retention Classification Calibration Metrics

Horizon	Model Architecture	Test Count	Empirical Coverage (%)	Conformal Quantile (q)	Avg Set Length (Labels)
<i>Standard Split Conformal Prediction (Global)</i>					
H=1	Logistic Regression	10,088	89.65%	0.684	1.406
H=1	MLP Classifier	10,088	89.79%	0.651	1.386
H=1	XGBoost Classifier	10,088	89.72%	0.663	1.384
H=2	Logistic Regression	9,938	89.54%	0.643	1.540
H=2	MLP Classifier	9,938	88.95%	0.636	1.509
H=2	XGBoost Classifier	9,938	89.03%	0.631	1.505
H=3	Logistic Regression	9,799	89.38%	0.638	1.554
H=3	MLP Classifier	9,799	89.08%	0.651	1.539
H=3	XGBoost Classifier	9,799	88.97%	0.636	1.534
H=4	Logistic Regression	9,651	88.51%	0.651	1.522
H=4	MLP Classifier	9,651	88.18%	0.657	1.497
H=4	XGBoost Classifier	9,651	88.04%	0.651	1.495
H=5	Logistic Regression	9,517	88.39%	0.670	1.472
H=5	MLP Classifier	9,517	87.90%	0.679	1.456
H=5	XGBoost Classifier	9,517	88.08%	0.677	1.454
<i>Locally Adaptive Conformal Prediction (LACP / Mondrian)</i>					
H=1	Logistic Regression	10,088	89.68%	1.7483	1.405
H=1	MLP Classifier	10,050	89.95%	1.9356	1.404
H=1	XGBoost Classifier	10,073	89.72%	1.9567	1.394
H=2	Logistic Regression	9,875	90.35%	1.5218	1.560
H=2	MLP Classifier	9,814	90.94%	1.6918	1.575
H=2	XGBoost Classifier	9,854	90.41%	1.6056	1.560
H=3	Logistic Regression	9,494	92.09%	1.5921	1.634
H=3	MLP Classifier	9,548	92.45%	1.7689	1.659
H=3	XGBoost Classifier	9,599	91.33%	1.6429	1.613
H=4	Logistic Regression	9,425	91.60%	1.9218	1.614
H=4	MLP Classifier	9,459	91.48%	2.0914	1.608
H=4	XGBoost Classifier	9,502	90.92%	1.9467	1.591
H=5	Logistic Regression	9,380	90.70%	2.1753	1.551
H=5	MLP Classifier	9,377	90.88%	2.2824	1.550
H=5	XGBoost Classifier	9,410	90.34%	2.1677	1.532

Table 2: Model 2 Active Retention classification coverage, quantiles, and average prediction set lengths across forecasting horizons ($H = 1 \dots 5$).

Table A3: Corbin Carroll Multi-Year Simulation Trajectory (Split Conformal)

Year	Age	Active Rate (%)	Set Len	Pred. WAR	90% Conformal Interval	Width	Accumulated WAR
<i>XGBoost Model</i>							
Year 0	21	100.0%	0	+0.00	[0.00]	0.00	0.00
Year 1	22	100.0%	1	+1.40	[1.40]	0.00	1.40
Year 2	23	100.0%	1	+5.40	[5.40]	0.00	6.80
Year 3	24	99.3%	1	+5.10	[+3.15, +7.05]	3.90	11.90
Year 4	25	98.8%	1	+4.33	[+2.15, +6.51]	4.35	16.23
Year 5	26	98.6%	1	+5.14	[+2.74, +7.54]	4.80	21.37
Year 6	27	98.2%	1	+3.86	[+1.42, +6.31]	4.89	25.23
Year 7	28	98.2%	1	+4.32	[+1.74, +6.90]	5.16	29.55
Year 8	29	99.6%	1	+3.60	[+1.65, +5.55]	3.90	33.15
Year 9	30	99.0%	1	+3.02	[+0.84, +5.19]	4.35	36.17
Year 10	31	95.5%	1	+3.02	[+0.62, +5.42]	4.80	39.19
Year 11	32	91.6%	1	+2.57	[+0.13, +5.02]	4.89	41.76
Year 12	33	84.9%	1	+2.10	[-0.48, +4.68]	5.16	43.86
Year 13	34	95.7%	1	+1.63	[-0.32, +3.58]	3.90	45.49
Year 14	35	81.7%	1	+1.16	[-1.02, +3.34]	4.35	46.65
Year 15	36	62.3%	2	+1.15	[-1.25, +3.55]	4.80	47.80
Year 16	37	41.3%	2	+0.85	[-1.59, +3.30]	4.89	48.65
<i>Linear/Logistic Model</i>							
Year 0	21	100.0%	0	+0.00	[0.00]	0.00	0.00
Year 1	22	100.0%	1	+1.40	[1.40]	0.00	1.40
Year 2	23	100.0%	1	+5.40	[5.40]	0.00	6.80
Year 3	24	99.7%	1	+4.08	[+2.12, +6.04]	3.92	10.88
Year 4	25	99.4%	1	+4.15	[+1.91, +6.39]	4.47	15.03
Year 5	26	98.6%	1	+4.26	[+1.86, +6.67]	4.81	19.29
Year 6	27	97.3%	1	+4.02	[+1.53, +6.52]	4.98	23.31
Year 7	28	95.6%	1	+4.02	[+1.35, +6.68]	5.33	27.33
Year 8	29	99.5%	1	+3.42	[+1.46, +5.38]	3.92	30.75
Year 9	30	97.7%	1	+3.15	[+0.91, +5.38]	4.47	33.90
Year 10	31	94.0%	1	+2.92	[+0.51, +5.32]	4.81	36.82
Year 11	32	88.2%	1	+2.69	[+0.20, +5.18]	4.98	39.51
Year 12	33	79.8%	1	+2.49	[-0.17, +5.16]	5.33	42.00
Year 13	34	96.7%	1	+2.02	[+0.06, +3.98]	3.92	44.02
Year 14	35	87.3%	1	+1.74	[-0.49, +3.98]	4.47	45.76
Year 15	36	72.3%	1	+1.51	[-0.90, +3.91]	4.81	47.27
Year 16	37	54.8%	2	+1.29	[-1.21, +3.78]	4.98	48.56
Year 17	38	37.7%	2	+1.08	[-1.59, +3.74]	5.33	49.64
Year 18	39	81.5%	1	+0.65	[-1.31, +2.61]	3.92	50.29
Year 19	40	54.0%	2	+0.36	[-1.88, +2.59]	4.47	50.65
<i>Neural Network Model</i>							
Year 0	21	100.0%	0	+0.00	[0.00]	0.00	0.00
Year 1	22	100.0%	1	+1.40	[1.40]	0.00	1.40
Year 2	23	100.0%	1	+5.40	[5.40]	0.00	6.80
Year 3	24	99.9%	1	+4.26	[+2.33, +6.20]	3.87	11.06
Year 4	25	99.8%	1	+4.20	[+1.99, +6.41]	4.42	15.26
Year 5	26	99.2%	1	+4.27	[+1.84, +6.69]	4.86	19.53
Year 6	27	98.6%	1	+3.92	[+1.44, +6.41]	4.97	23.45
Year 7	28	95.1%	1	+3.30	[+0.69, +5.91]	5.22	26.75
Year 8	29	99.7%	1	+2.90	[+0.97, +4.84]	3.87	29.65
Year 9	30	98.6%	1	+2.79	[+0.58, +5.00]	4.42	32.44
Year 10	31	95.1%	1	+2.39	[-0.04, +4.82]	4.86	34.83
Year 11	32	89.9%	1	+2.08	[-0.40, +4.57]	4.97	36.91
Year 12	33	84.0%	1	+2.01	[-0.60, +4.62]	5.22	38.92
Year 13	34	97.4%	1	+1.31	[-0.63, +3.24]	3.87	40.23
Year 14	35	80.1%	1	+1.48	[-0.73, +3.69]	4.42	41.71
Year 15	36	65.2%	2	+1.26	[-1.17, +3.69]	4.86	42.97
Year 16	37	42.7%	2	+0.94	[-1.54, +3.43]	4.97	43.91

Table 3: Corbin Carroll year-by-year projected WAR, 90% split conformal prediction intervals, active retention rates, set lengths, and cumulative accumulated WAR.

Table A4: Corbin Carroll Multi-Year Simulation Trajectory (Locally Adaptive Conformal)

Year	Age	Active Rate (%)	Set Len	Pred. WAR	90% Conformal Interval	Width	Accumulated WAR
<i>XGBoost Model</i>							
Year 0	21	100.0%	1	+0.00	[0.00]	0.00	0.00
Year 1	22	100.0%	1	+1.40	[1.40]	0.00	1.40
Year 2	23	100.0%	1	+5.40	[5.40]	0.00	6.80
Year 3	24	99.3%	1	+5.10	[+3.33, +6.87]	3.53	11.90
Year 4	25	98.8%	1	+4.33	[+2.02, +6.64]	4.61	16.23
Year 5	26	98.6%	1	+5.14	[+2.93, +7.34]	4.40	21.37
Year 6	27	98.2%	1	+3.86	[+1.09, +6.63]	5.54	25.23
Year 7	28	98.2%	1	+4.32	[+1.58, +7.07]	5.49	29.55
Year 8	29	99.6%	1	+3.60	[+1.52, +5.68]	4.16	33.15
Year 9	30	99.0%	1	+3.02	[+0.80, +5.23]	4.42	36.17
Year 10	31	95.5%	1	+3.02	[+0.58, +5.46]	4.88	39.19
Year 11	32	91.6%	1	+2.57	[+0.13, +5.02]	4.89	41.76
Year 12	33	84.9%	1	+2.10	[-0.42, +4.62]	5.04	43.86
Year 13	34	95.7%	1	+1.63	[-0.45, +3.71]	4.16	45.49
Year 14	35	81.7%	1	+1.16	[-1.05, +3.37]	4.42	46.65
Year 15	36	62.3%	2	+1.15	[-1.28, +3.59]	4.88	47.80
Year 16	37	41.3%	2	+0.85	[-1.59, +3.30]	4.89	48.65
<i>Linear/Logistic Model</i>							
Year 0	21	100.0%	1	+0.00	[0.00]	0.00	0.00
Year 1	22	100.0%	1	+1.40	[1.40]	0.00	1.40
Year 2	23	100.0%	1	+5.40	[5.40]	0.00	6.80
Year 3	24	99.7%	1	+4.08	[+2.31, +5.86]	3.55	10.88
Year 4	25	99.4%	1	+4.15	[+1.78, +6.52]	4.73	15.03
Year 5	26	98.6%	1	+4.26	[+1.99, +6.53]	4.54	19.29
Year 6	27	97.3%	1	+4.02	[+1.27, +6.77]	5.50	23.31
Year 7	28	95.6%	1	+4.02	[+1.26, +6.77]	5.51	27.33
Year 8	29	99.5%	1	+3.42	[+1.34, +5.50]	4.16	30.75
Year 9	30	97.7%	1	+3.15	[+0.89, +5.41]	4.53	33.90
Year 10	31	94.0%	1	+2.92	[+0.47, +5.36]	4.89	36.82
Year 11	32	88.2%	1	+2.69	[+0.20, +5.18]	4.98	39.51
Year 12	33	79.8%	1	+2.49	[-0.10, +5.09]	5.19	42.00
Year 13	34	96.7%	1	+2.02	[-0.06, +4.10]	4.16	44.02
Year 14	35	87.3%	1	+1.74	[-0.52, +4.01]	4.53	45.76
Year 15	36	72.3%	1	+1.51	[-0.94, +3.95]	4.89	47.27
Year 16	37	54.8%	2	+1.29	[-1.21, +3.78]	4.98	48.56
Year 17	38	37.7%	2	+1.08	[-1.52, +3.67]	5.19	49.64
Year 18	39	81.5%	1	+0.65	[-1.43, +2.73]	4.16	50.29
Year 19	40	54.0%	2	+0.36	[-1.91, +2.62]	4.53	50.65
<i>Neural Network Model</i>							
Year 0	21	100.0%	1	+0.00	[0.00]	0.00	0.00
Year 1	22	100.0%	1	+1.40	[1.40]	0.00	1.40
Year 2	23	100.0%	1	+5.40	[5.40]	0.00	6.80
Year 3	24	99.9%	1	+4.26	[+2.48, +6.05]	3.56	11.06
Year 4	25	99.8%	1	+4.20	[+1.90, +6.50]	4.59	15.26
Year 5	26	99.2%	1	+4.27	[+2.05, +6.48]	4.43	19.53
Year 6	27	98.6%	1	+3.92	[+1.17, +6.68]	5.50	23.45
Year 7	28	95.1%	1	+3.30	[+0.54, +6.06]	5.52	26.75
Year 8	29	99.7%	1	+2.90	[+0.83, +4.98]	4.15	29.65
Year 9	30	98.6%	1	+2.79	[+0.54, +5.04]	4.50	32.44
Year 10	31	95.1%	1	+2.39	[-0.10, +4.88]	4.98	34.83
Year 11	32	89.9%	1	+2.08	[-0.44, +4.61]	5.06	36.91
Year 12	33	84.0%	1	+2.01	[-0.52, +4.54]	5.06	38.92
Year 13	34	97.4%	1	+1.31	[-0.77, +3.38]	4.15	40.23
Year 14	35	80.1%	1	+1.48	[-0.77, +3.73]	4.50	41.71
Year 15	36	65.2%	2	+1.26	[-1.17, +3.69]	4.86	42.97
Year 16	37	42.7%	2	+0.94	[-1.54, +3.43]	4.97	43.91
Year 17	38	28.5%	2	+0.79	[-1.74, +3.32]	5.06	44.70
Year 18	39	88.9%	1	+0.70	[-1.37, +2.78]	4.15	45.40
Year 19	40	56.4%	2	+0.45	[-1.80, +2.69]	4.50	45.85
Year 20	41	44.9%	2	+0.33	[-2.16, +2.82]	4.98	46.18

Table 4: Corbin Carroll year-by-year projected WAR, 90% locally adaptive (Mondrian) conformal prediction intervals, active retention rates, set lengths, and cumulative accumulated WAR.

Table A5: Top 10 Historical Peer Group Comparisons

Top 10 Overall Match (CCTD)				Top 10 Structural Comps (Mahalanobis)				Top 10 Closest Career WAR			
Name	Mahal.	WAR	CCTD	Name	Mahal.	WAR	CCTD	Name	Mahal.	WAR	CCTD
Jose Ramirez	2.83	44.9	0.225	Tommy Edman	2.37	14.5	0.505	Andrew McCutchen	4.02	51.9	0.320
Jose Reyes	2.89	43.9	0.233	Jazz Chisholm Jr.	2.40	5.4	0.627	David Wright	5.28	51.2	0.416
Brett Gardner	3.10	38.5	0.276	Michael Harris II	2.47	6.6	0.613	Jose Altuve	3.60	49.7	0.283
Trea Turner	2.47	33.3	0.281	Trea Turner	2.47	33.3	0.281	Jimmy Rollins	4.26	49.6	0.335
Jose Altuve	3.60	49.7	0.283	Jarrod Dyson	2.57	13.6	0.523	Ian Kinsler	4.37	47.4	0.342
Francisco Lindor	3.60	45.2	0.284	Bobby Witt Jr.	2.59	4.9	0.639	Curtis Granderson	3.91	47.2	0.306
Mookie Betts	3.50	54.7	0.291	Cedric Mullins II	2.61	11.6	0.550	Francisco Lindor	3.60	45.2	0.284
Ryan Braun	3.66	43.6	0.292	Tim Lincecum	2.63	1.2	0.689	Bryce Harper	4.31	45.0	0.339
Starling Marte	2.97	34.5	0.299	Wander Franco	2.75	8.3	0.598	Jose Ramirez	2.83	44.9	0.225
Curtis Granderson	3.91	47.2	0.306	Daz Cameron	2.80	-0.2	0.712	Jose Reyes	2.89	43.9	0.233

Table 5: Top 10 historical peer group comparisons ranked by Overall CCTD, Structural Mahalanobis Distance, and Closest Career WAR Alignment.